



How would you feel, if you can control things just by clapping your hands like a king? Presented here is a circuit that lets you fulfil your wish. The circuit can switch two or more devices 'on' and 'off' in response to a series of rapid handclaps.

Fig. 1 shows the circuit of a clap-controlled switch. The circuit is built around two NE555 timers (IC2 and IC3), decade counter 4017B (IC4), operational amplifier LM741 (IC1), dual J-K flip-flop CD4027 (IC5) and a few discrete components.

The claps are picked up by electret microphone MIC and amplified by op-amp IC1. IC1 is configured as an

inverting amplifier, with its gain and hence the sensitivity of the circuit adjustable via trimpot VR1. Its output is then fed into a pulse-shaping stage based on 555 timer IC2, which is configured as a monostable multivibrator with its trigger input (pin 2) normally biased above $1/3V_{cc}$ using resistors R5 and R6.

When a loud enough sound is detected, pin 2 is pulled below $1/3V_{cc}$, triggering IC2. The output (pin 3) of IC2 immediately goes high, causing transistor T1 to conduct. The result is approximately 250ms low-going pulse at the collector of T1.

The output from the pulse-shaper stage is fed to a missing pulse detector based on 555 timer IC3, which is also configured as a monostable multivibrator. However, this monostable

circuit differs from monostable IC2 as it includes a retrigger function.

Retriggering is made possible

Test point	Details
TP0	0V, GND
TP1	6-12V depending on the input supply
TP2	Low-going pulse at each clap
TP3	High for 250 ms (approx.) at each clap
TP4	Low for 250 ms (approx.) at each clap
TP5	High for 750 ms (approx.) at each clap
TP6	High to low after 750 ms (approx.) once triggered

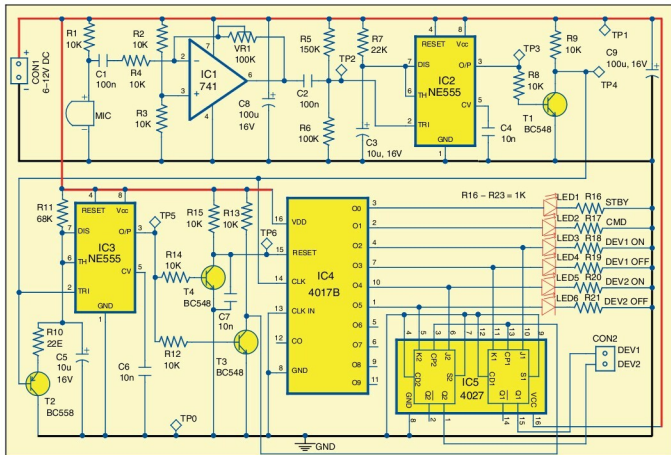


Fig. 1: Circuit of clap switch